

Privacy-Preserving Machine Learning

An Automated, Data-Driven Approach to Parameter Selection for Homomorphic Encryption

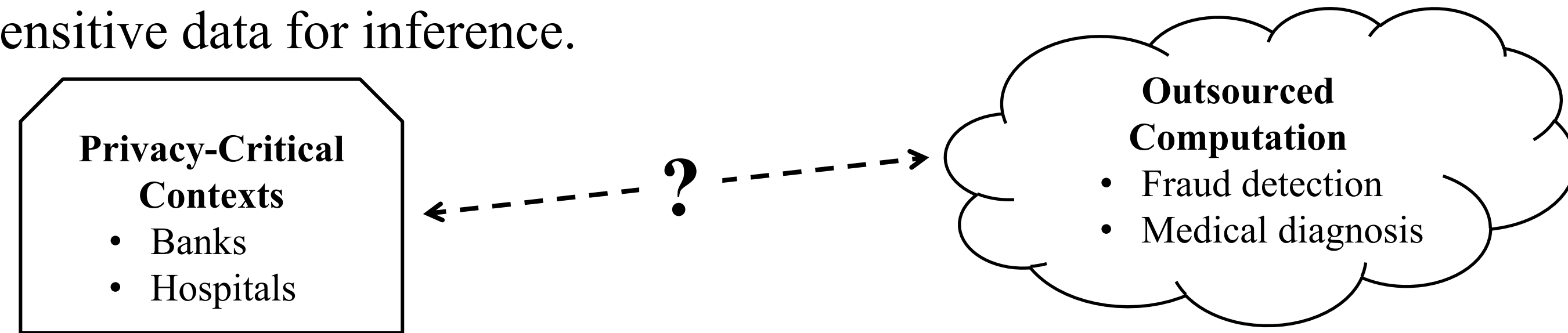
Jonathan Rueffer | Computer Science | Advised by Alex Nord

Abstract

Homomorphic encryption enables computation directly on encrypted data, making it promising for privacy-critical machine learning applications such as fraud detection and medical diagnosis. The CKKS scheme is particularly well-suited for inference on real-valued data, but its practical use is limited by the difficulty of selecting parameters that balance security, precision, and efficiency. This work presents an automated, data-driven framework for CKKS parameter selection that incorporates dataset and model characteristics to generate and evaluate candidate configurations. Using representative test vectors instead of sensitive data, the framework supports a multi-objective trade-off analysis of performance metrics.

Context

- Machine learning models offer powerful predictive capabilities but are often hosted by third-party cloud providers, requiring users to submit sensitive data for inference.



→ How can outsourced models be leveraged in privacy-critical settings?

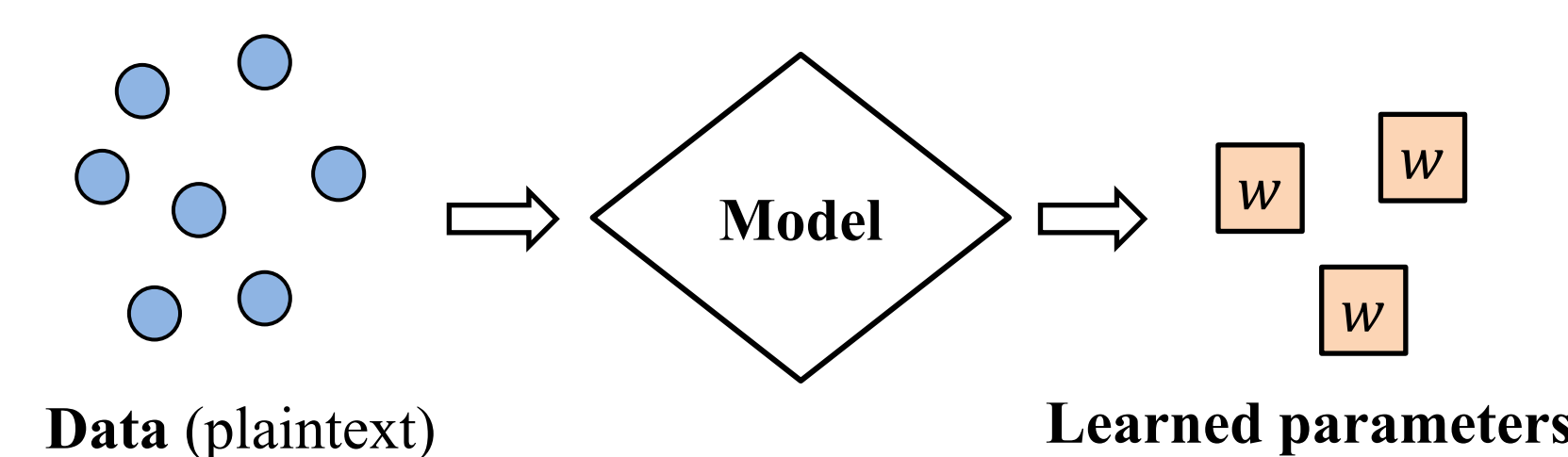
Homomorphic Encryption

- Homomorphic encryption enables computation directly on encrypted data. Supported operations (primarily addition and multiplication) preserve the underlying message without decryption.
- In machine learning, this requires models to be expressed using only these operations (e.g., polynomial approximations of non-linear activation functions).
- Under a public-key homomorphic encryption scheme, a model f can perform inference on encrypted data:

$$\text{Dec}_{sk} \left(f \left(\text{Enc}_{pk}(m) \right) \right) = f(m)$$

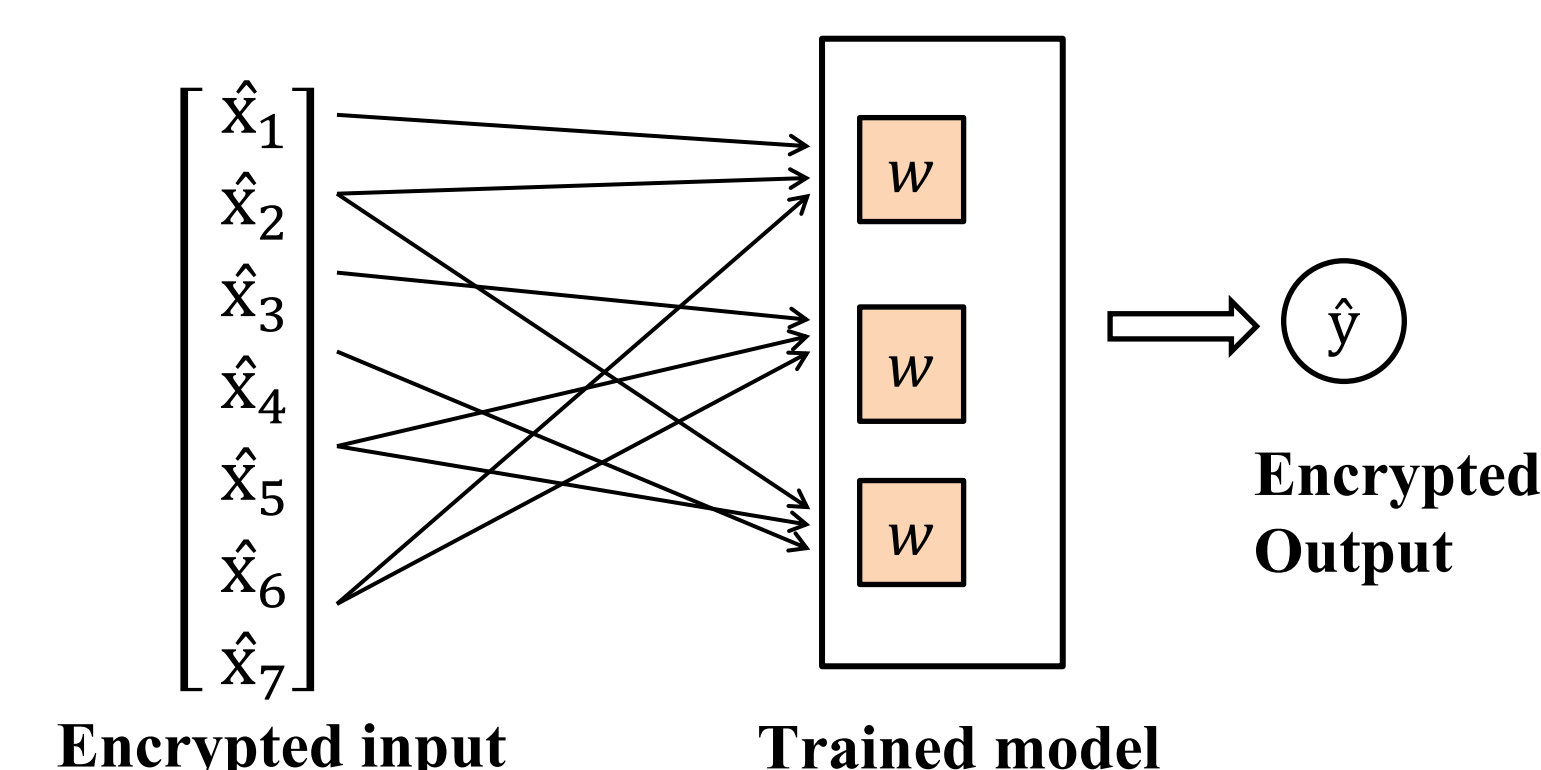
Model Training

- The cloud-based model is trained on unencrypted data, learning patterns that are stored as fixed parameters (e.g., weights).



Model Inference

- During encrypted inference, the model applies the same deterministic computation to encrypted inputs using the learned parameters, producing the same result after decryption.

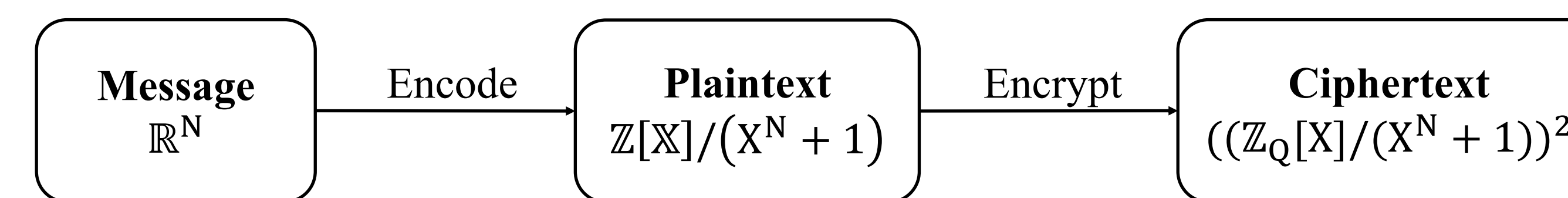


The CKKS Scheme

- The Cheon–Kim–Kim–Song (CKKS) scheme is well-suited for privacy-preserving machine learning, as it supports encrypted real-valued data and approximate arithmetic.

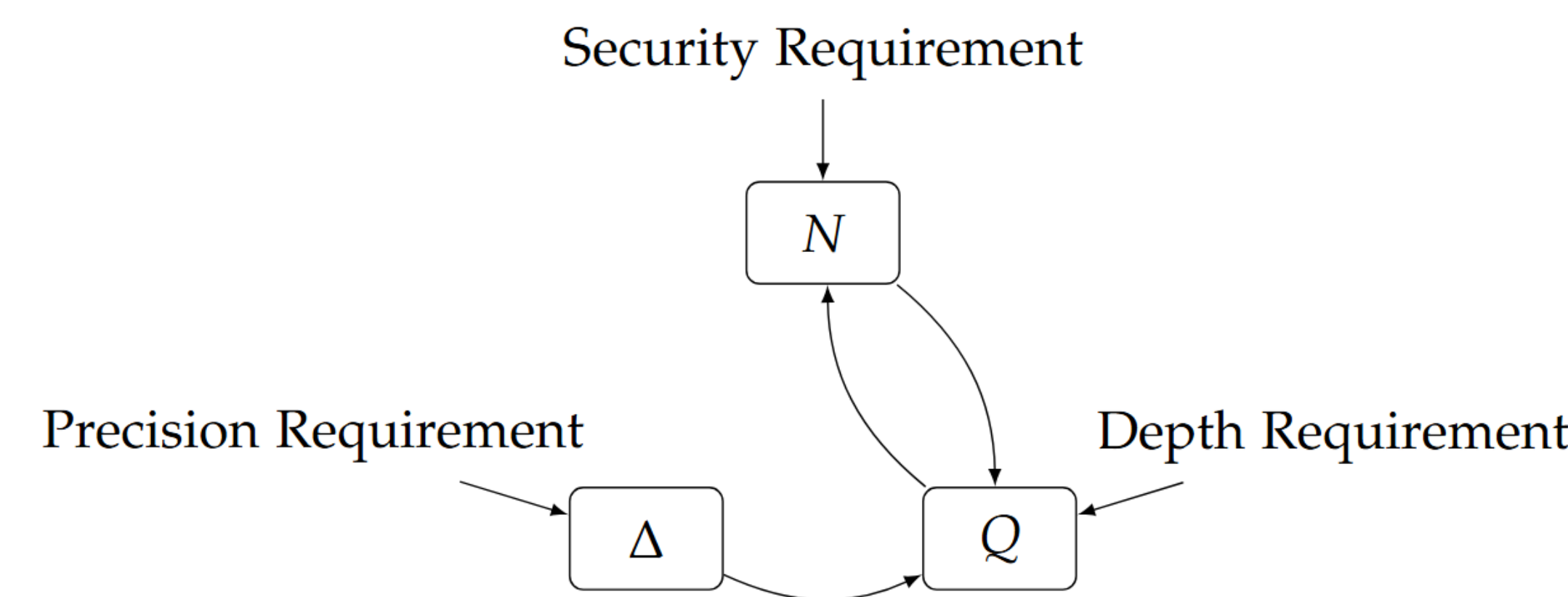
$$\text{Dec}(\text{Enc}(x) + \text{Enc}(y)) \approx x + y$$

- Security is based on the hardness of the Ring Learning With Errors (RLWE) problem, making it quantum-resistant.
- Correctness requires managing small noise introduced during encryption and computation.



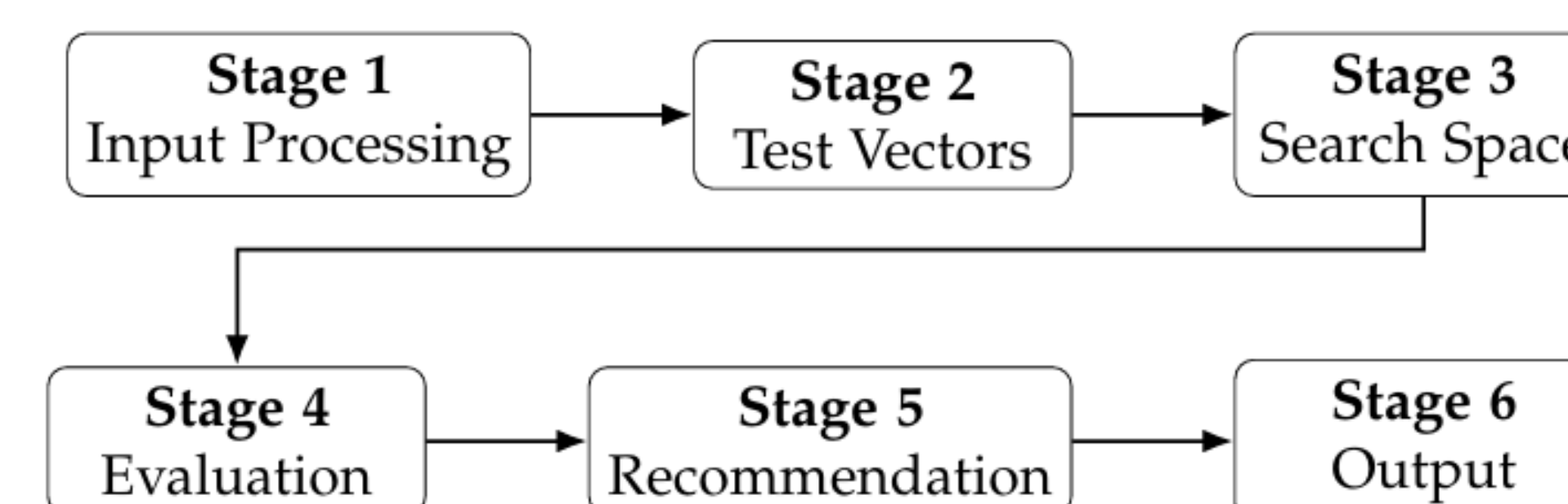
CKKS Parameters

- Δ (scale):** Scales real-valued inputs before encoding into integers, controlling precision. Larger scales preserve more decimal bits of precision.
- N (polynomial modulus degree):** Determines the size of the modulus polynomial $(X^N + 1)$. Larger N increases capacity and security.
- Q (coefficient modulus):** Defines ciphertext coefficient size. In practice, $Q = q_0 q_1 \dots q_L$, where each ciphertext-ciphertext multiplication consumes one level q_i to maintain precision.



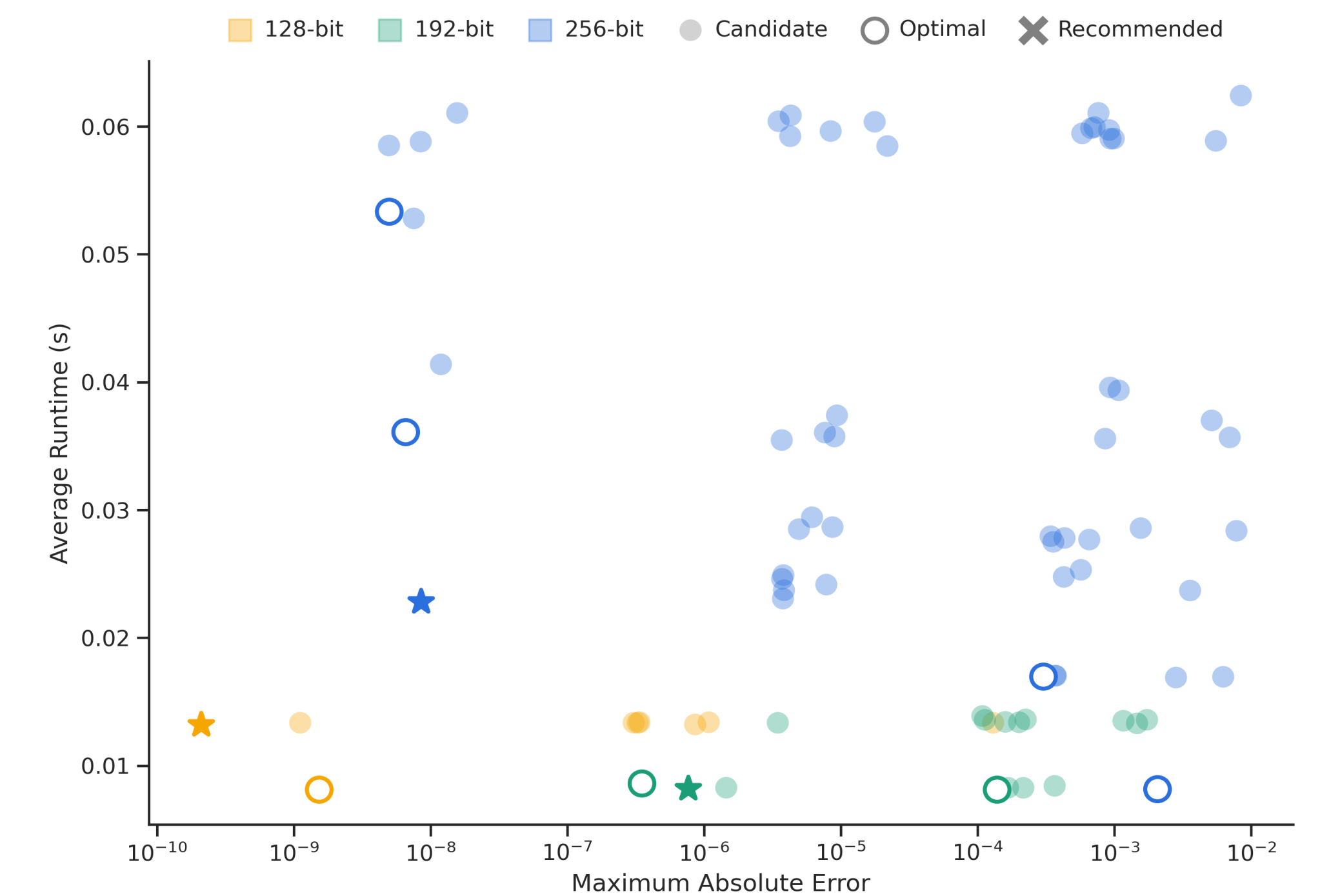
Automated and Data-Driven CKKS Parameter Selection

- Selecting CKKS parameters is complex, with choices affecting security, precision, and computational cost. To reduce the expertise required, I developed a tool that automates this process.
- Key contributions:**
 - Incorporates model and dataset constraints into parameter selection
 - Securely evaluates candidate parameter configurations
 - Frames selection as a trade-off between precision, security, and performance



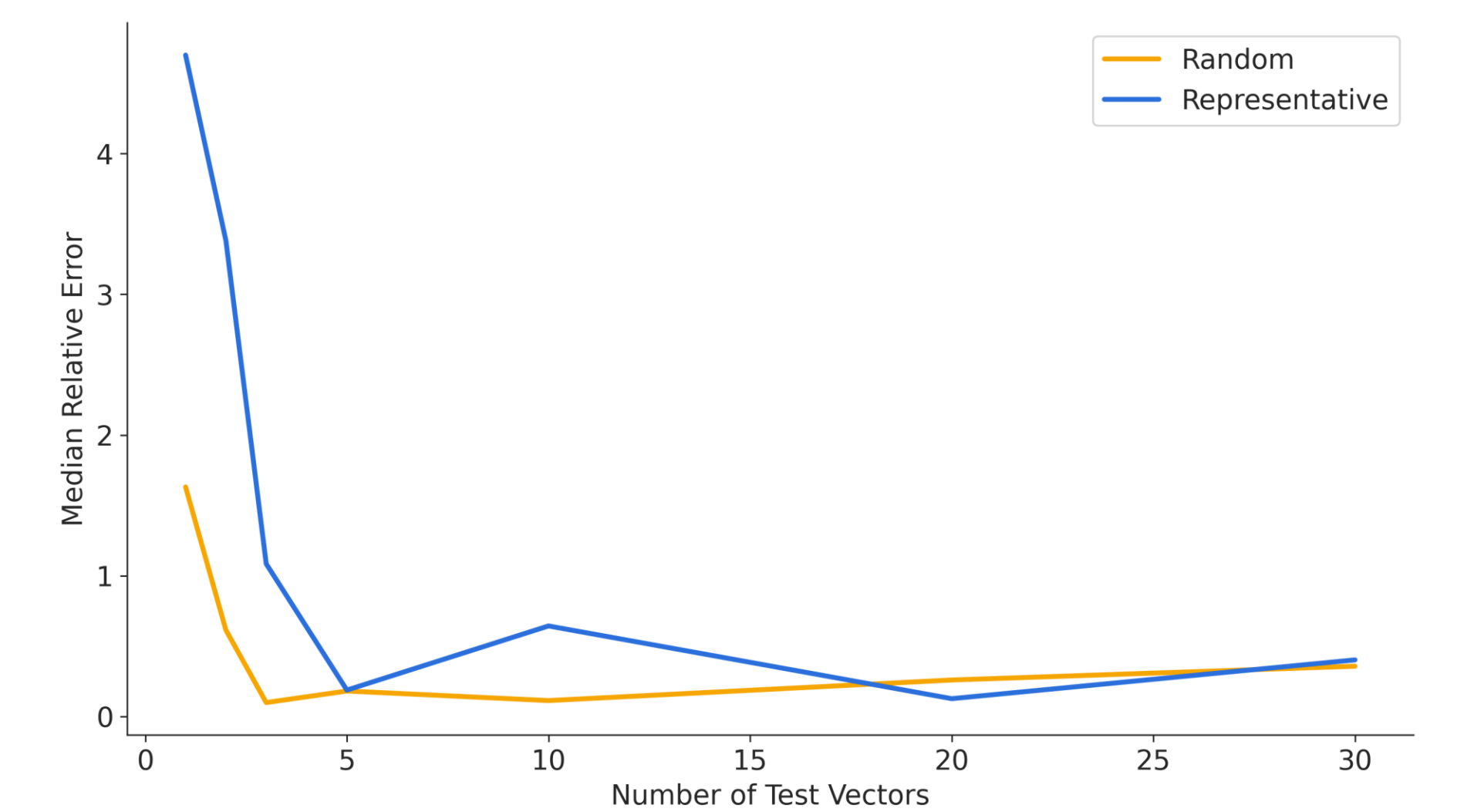
Results

Healthcare Case Study: Parameter Selection as a Trade-off Analysis



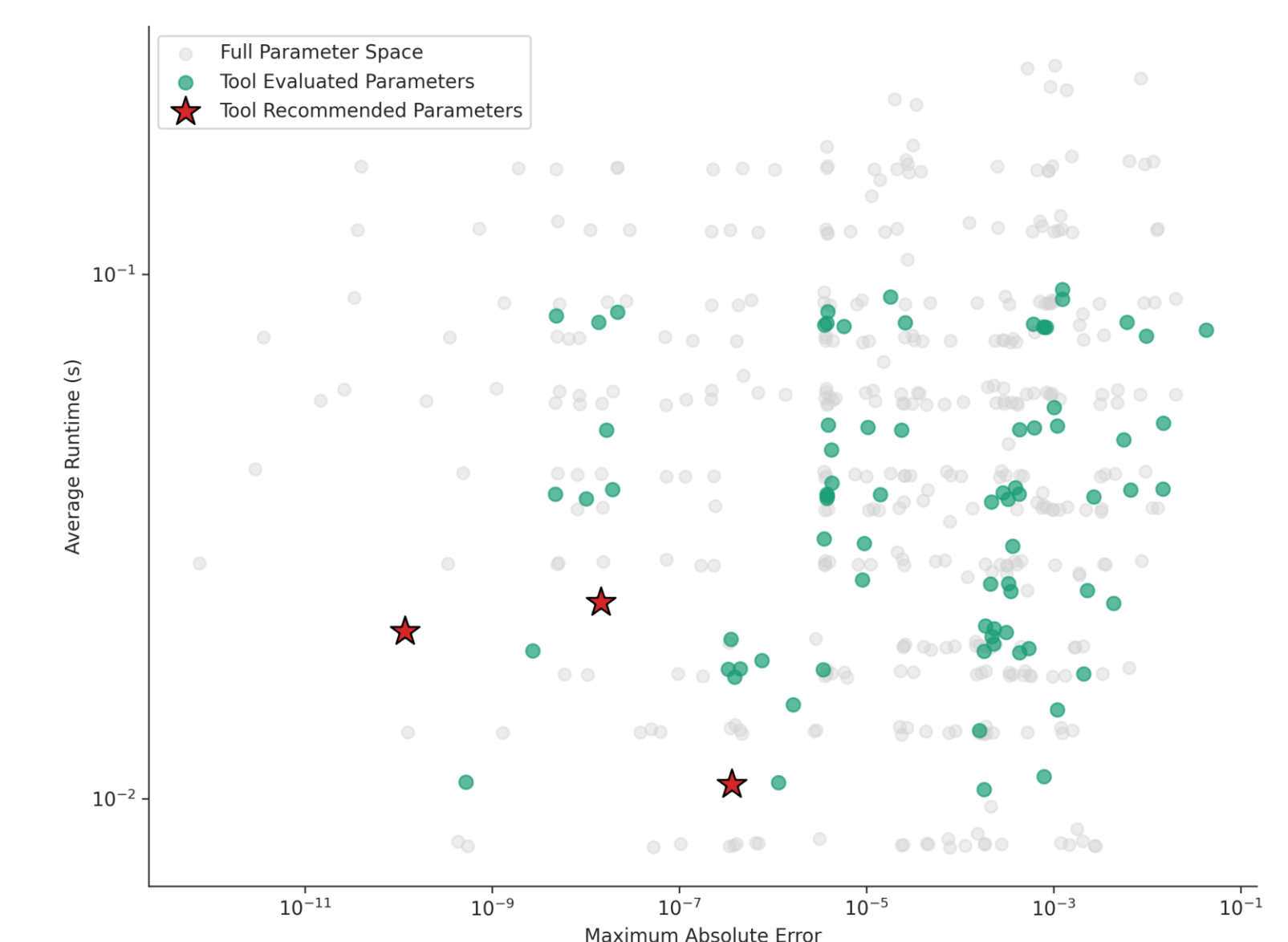
- The tool evaluates candidate parameter sets across security level, runtime, and error. It identifies the Pareto frontier of optimal runtime–error trade-offs and selects a single practical configuration as the point closest to the origin for usability.

Effectiveness of Representative Test Vectors



- Representative test vectors are synthetic inputs generated from the statistical properties of the dataset, avoiding the need to use real plaintext data for evaluation. Error convergence demonstrates effective approximation of dataset behavior.

Effectiveness of Data-Informed Parameter Selection



- The tool constructs a restricted search space using dataset and model properties (e.g., data resolution and multiplicative depth). It produces configurations near the optimal region of an exhaustive search while significantly reducing cost (10 → 1.5 minute runtime; 380 → 80 sets).